Smart Coastline Environment Management Using Deep Detection of Manmade Pollution and Hazards

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***Abstract*—A significant portion of the tourism industry and the fishing communities depend on the health of the seashores for their livelihood. Visual interpretation of this environment provides the required information for autonomous agents for decision support in large-scale operations such as smart beach management. Deep Neural Networks (DNN) are recently used for image classification and scene understanding with very good results. In this paper, a DNN is used for object detection in the scenes from seashore areas. This interpretation is used for decision support for various smart beach applications. The technique used for smart beach scene interpretation is the transfer learning on a pre-trained DNN using VGG architecture. The Single Shot Detector (SSD) technique is used for object detection in the collected dataset from the beach areas. A dataset from the beaches of the Caspian Sea is collected in order to provide an extensive simulation in the real-world setting. The experimental results showed that the accuracy of the presented technique is acceptable for various applications in the seashore environment.**

Keywords—Deep Learning, Computer Vision, Smart Beach, Object Detection.

I. Introduction

The coastline of the oceans, seas and lakes have major ecological, financial and social roles for many communities around the world. The environmental impact of the human intervention in the seashores has negative influence on the ecology of marine life. The pollution resulted from tourism and fishing leaves many beaches around the world in critical environmental conditions. In recent years, smart environments provide a solution for optimized management of the cities and countries. A collection of sensor networks, robotic agents and information technology based management give us the ability to model the complex decisions requires for managing the large human communities. In this paper the smart city model is extended to seashore areas and from now on this model is referred to as the Smart Beach. Visual analysis is one of the most important tools for smart environment management. Videos provided by the fixed surveillance cameras or cameras mounted on deployable robotic agents provides the management software system in the smart environment with necessary information for various complex decisions [1- 4].

Deep Neural Networks (DNN) are recently provided accurate solutions for various complex computer vision applications. Lu et al. [4] proposed an efficient deep learning model for object detection in robotic applications. Liu et al. [5] provided a survey of the recent advances in DNN and their applications. Girshick et al. [6] proposed the region based convolutional neural networks for accurate object detection and segmentation. Ren et al. [7] proposed fast convolutional neural networks for real time object detection. The application of predictive methods and especially neural networks lead to good results for smart environment applications [8- 12].

In this paper the DNN are used for interpretation of the scenes in the beach areas. The technique used for object detection uses the transfer learning on VGG architecture [13] to interpret the videos for various smart beach applications. For simulation, a large data set from the coastline of Caspian Sea is collected. The experimental results on various smart beach applications show that the proposed model have acceptable accuracy in different scenarios. The rest of the paper is organized as follows: Section II describes the proposed deep neural network framework for object detection. The experimental results and the simulations are described in Section III and finally Section IV concludes the paper.

II. Deep Learning Based Object Detection in Seashore Scenes

In the deep learning framework used for object detection in the seashore scenes, the VGG deep convolutional neural network is used. The framework for implementation of the deep network is the Caffe framework developed by the University of Berkeley [14]. The architecture for VGG deep neural network is presented in Figure 1.

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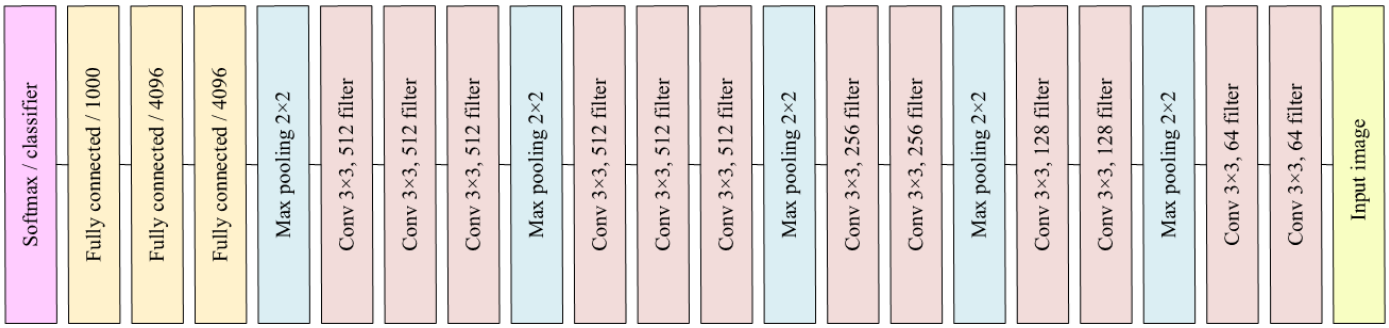

Fig. 1. The architecture for VGG deep neural network

As shown in Figure 1 the VGG-16 architecture uses 12 convolutional layers followed by an average pooling layer, fully connected layers and a Softmax layer. In Comparison with the AlexNet, the VGG is more efficient deep neural network, because it reduces the number of trainable parameters by using stack of convolutional layers with filter size 3×3 instead of using bigger filter size. This is due to the fact that the stack of three convolutional layers with filter size of 3×3 and stride 1, has the same effective receptive field as one convolutional layer with filter size of 7×7. But the number of parameters in stack of 3×3 convolution layers are $3(3^2C^2)$ and the number of parameters in single 7×7 convolution layer are 7^2C^2, where C is the number of channels per layer, so the network would be deeper but it has fewer parameters.

The total number of 24 object classes are used for training the deep neural network. The used object classes are the same as MSCOCO image dataset which includes background class, airplanes, bicycles, birds, boats, bottles, buses, cars, cats, chairs, cows, dining tables, dogs, horses, motorbikes, people, potted plants, sheeps, sofas, trains, and TV monitors.

One of the main goals of the proposed deep model was fast and accurate interpretation of the coastline environment with the low computational capacity of the embedded processing unit of a robotic agent. The VGG model for scene interpretation has a section of the neural network dedicated to proposing a region and the next section is a classifier for classification of these proposals. This model has high accuracy but the computational complexity results in a low frame rate. A method for solving this problem is localizing the convolutional filters. In order to perform this task, the proposal section of the network is combined with the classification part using a group of predefined boxes.

The features produced using the convolutional filters in the later layers of the network are used for producing the class probability and the place of the bounding box using a small convolutional filter. In order to have the location of the bounding boxes with lower computational complexity, for localizing the objects in the image the later convolutional layers are used which have smaller image size due to the effect of the final layers of the neural network. In the later layers of the convolutional network each pixel contains a bigger section of the input image and therefore calculation of the object positions are less time-consuming. This model is called the Single Shot Detector (SSD) [15] method. The SSD method uses the bounding box and ground truth for the objects in the image in order to train the VGG deep network. The SSD is the extended multi-object version of the Multibox objective. The loss function of the SSD is the weighted sum of the localization loss and the confidence loss as follows:

$$L(x,c,l,g) = \frac{1}{N}\left(L_{conf}(x,c) + \alpha L_{loc}(x,l,g)\right) \tag{1}$$

The confidence loss is defined as the loss for the multi class Softmax as follows:

$$L_{conf}(x,c) = -\sum_{i\in Pos} x_{ij}^{p}\, log\left(\hat{c}_i^p\right) - \sum_{i\in Neg} log\left(\hat{c}_i^0\right) \tag{2}$$

in which

$$\hat{c}_i^p = \frac{exp(c_i^p)}{\sum_p exp(c_i^p)} \tag{3}$$

III. Experimental Results

In this paper, a dataset for smart beach applications is gathered in Gilan province of Iran. There is a collection of videos with the overall size of 20GB in this dataset and the resolution of the videos are 1920 × 1080 pixels. A sample of the collected smart beach dataset video frames are presented in Figure 2.

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Fig. 2. Sample videoframes from smart beach dataset.

In order to simulate the smart beach environment management two scenarios are simulated. The first scenario is detection of illegal entry by vehicles and boats into the public beach area. The second scenario is the detection of the illegal dumping of garbage in the beach areas.

The general model for vehicles and boats are present in the pre-trained VGG model. Further training for increasing accuracy is performed using local vehicles and boat models. As the visual model for garbage was not present in the pre-trained model a dataset of manmade pollution is collected and used for further training of the model. This dataset includes four categories of manmade pollution including plastic bags, glass bottles, plastic bottle and metal cans. Samples of this dataset is presented in Figure 3.

Fig. 3. Sample manmade poluttion dataset

The simulations demonstrated that using high resolution input images lead to many false classifications. Therefore, the video frames are resized to 300×300 pixels. The next stage of preprocessing is mean subtraction with mean value equal to 0.0008 and finally the scaling process with standard deviation of 128 is performed on the input images.

For the first scenario the pre-trained VGG model with further training on local vehicle and boat models is used to detect illegal entering of the vehicles and boats into the beach area in the video dataset. For second scenario the last layer of the pre-trained VGG deep neural network is retrained on manmade pollution dataset and then it is fine-tuned on the MSCOCO dataset.

For evaluation of the accuracy of object detection the Intersection Over Union (IOU) method is used. In the IOU method, first the ground truth data is generated manually and then the accuracy of the detected objects are calculated based on the division of the intersection of the ground truth bounding box and the detected object over the union of the ground truth bounding box and the detected object. The model for IOU is presented in Figure 4.

The accuracy and loss of retrained model presented in figures 4 (a) and 4 (b) respectively. The overall accuracy is 0.92 which is an acceptable accuracy for smart beach applications. The experimental results for detection of illegal vehicles, detection of illegal boats and garbage detection in the beach area are presented in figures 6, 7 and 8 respectively.

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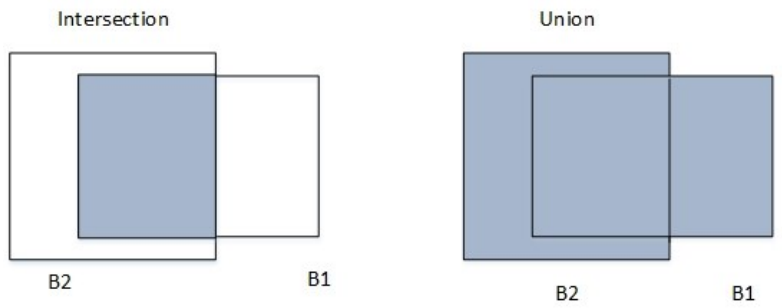
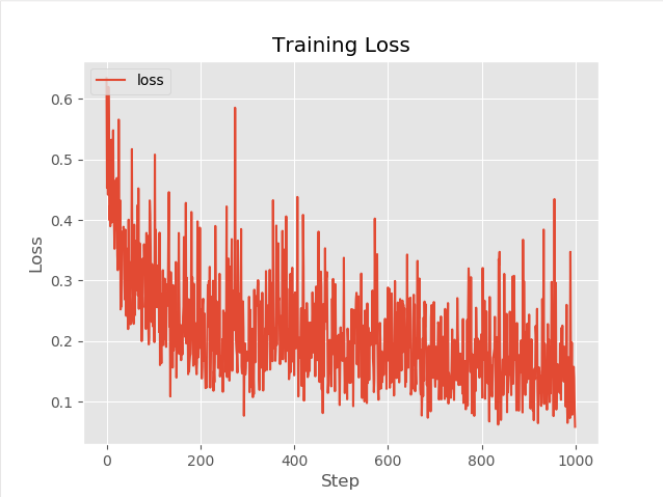


$$\mathbf{IoU} = \frac{\mathbf{B1} \cap \mathbf{B2}}{\mathbf{B1} \cup \mathbf{B2}}$$

Fig. 4. Intersection Over Union (IOU) method.



(a)

(b)

Fig. 5. (a) is the loss and (b) is the acuuracy of the trained model.

IV. Conclusion

In this paper a deep neural network based for autonomous management of the coastline areas is presented. The proposed technique uses the VGG architecture to semantically label the seashore scenes. The dataset from the various scenarios in beach areas is collected and a series of simulations are performed for various smart beach applications. The experimental results show that the trained model has acceptable accuracy and speed for visual interpretation of the seashore scenes.

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Fig. 6. Experimental results for detection of illegal vehicles in the beach area

Fig. 7. Experimental results for detection of illegal boats in the beach area

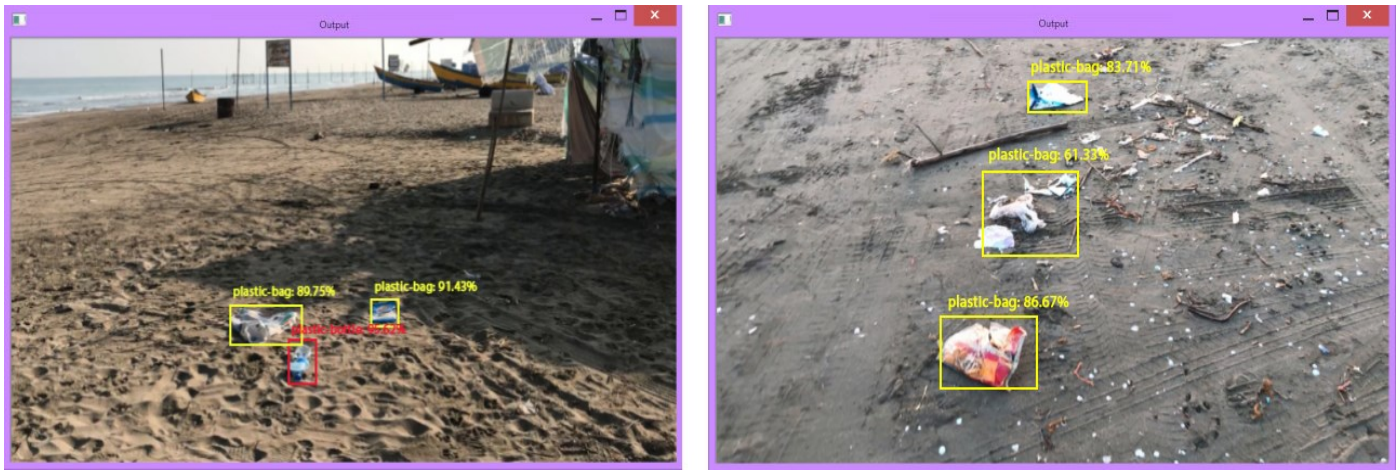

Fig. 8. Experimental results for detection of man made poluttion in the beach area

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